

Intermediate Elective

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Set up

Packages

```
# reads in tidyverse package from library
library(tidyverse)
# reads in janitor package from library
library(janitor)
# reads in here package from library
library(here)
# reads in naniar package from library
library(naniar)
# reads in patchwork package from library
library(patchwork)
# reads in viridis package from library
library(viridis)
# reads in scales package from library
library(scales)
```

Data

```
# read in weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))
# read in water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))
# read in water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

```
# reads in eBird data as birds
birds <- read_csv(here("data", "birds.csv"))
# read in Venco water data
venoco_water <- read_csv(here("data", "venoco-water.csv"))
```

1. Explain your inspiration

i Note

In 1-2 sentences each, describe:
which visualization you chose for your inspiration why it makes sense for your dataset of choice which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.)

Which visualization you chose for your inspiration: For my inspiration, I chose Steven Ponce's "Coal falls. Renewables rise." visualization because it uses a clean time-series design to show change over time which was visually appealing to me. I am applying that same idea to bird survey data by showing how observed bird counts change across water years. **Why it makes sense for your dataset of choice:** This inspiration makes sense for my dataset because the observations in the Ebird data are naturally time-based so a time-series figure can show whether selected species increased, decreased, or stayed stable over time. Instead of showing energy sources over time (like Steven Ponce's original does), my figure will show selected bird species observed at NCOS over time.

Which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.): The variables in shown in my figure will be **water_year** (found from original date column), **species**, and **count** (summed to **total_observations**). Water year will be on the x-axis, total observations will be on the y-axis, and species will determine line and point color color, and there will be labels to identify each species at the end of its line.

2. Plan your figure

i Note

Sketch what your figure would look like following your inspiration.
Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

i Note

Clean/wrangle your data as needed.

Then, code the figure you decided to create.

Make sure all your code is annotated, and that the output is showing.

At the end of this section, include code to save your final figure as a .jpg or .png file.

4. Describe your process

i Note

In 1-3 sentences each, describe:

your process of using someone else's code to create a visualization challenges you encountered as you made your visualization how your visualization differs from visualizations you made in class, for assignments, or for a previous elective